

CaP-Agent0 Direct Synthesis

Approach



Handle interaction



Final state



Simplified version of getting grasps

```
obs = get_observation()
grasps, scores = plan_grasp(obs, "cabinet handle")
best_idx = np.argmax(scores)
```

```
selected_grasp = grasps[best_idx] @ R_yawr
approach_dir = selected_grasp[:3, 2]
approach_pos = handle_pos - approach_dir * 0.15
goto_pose(approach_pos, quat)
goto_pose(handle_pos, quat)
close_gripper()
pull_dir = -approach_dir
goto_pose(handle_pos + pull_dir * 0.30, quat)
```

⊗ Failed

▲ Re-computes segmentation and grasp candidates at runtime.

▲ Derives approach vector and offset from low-level geometry.

▲ Manually constructs pull direction and travel distance.

Re-derives segmentation, grasp selection, approach geometry, and pull motion from lower-level operations.

Condensed policy code excerpts

RATS with learned-skills

Approach



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```
obs = get_observation()
grasp, pull_dir = plan_pull_grasp(obs, "cabinet handle")
```

if grasp is not None:

```
target_pos = localize_object_with_molmo_sam3("cabinet handle")
grasp_and_pull_horizontal_handle(target_pos, pull_dir)
```

✓ Success

▲ Plans pull grasp and pull direction using learned affordances. `plan_pull_grasp`

▲ Localizes the handle robustly with a learned vision model. `localize_object_with_molmo_sam3`

▲ Executes the pull with a learned handle-motion skill `grasp_and_pull_horizontal_handle`

Reuses localized pull-grasp planning and handle-motion abstractions.

Learned helper (RATS skill)